

Federico Cunico

PhD

Researcher in Computer Vision & AI · Software Engineer

PhD in Computer Science, University of Verona

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RESEARCH

I work on computer vision and deep learning for human-centered scene analysis — from industrial human-robot collaboration to robotic-assisted surgery. I am also a full-stack software engineer, comfortable from research prototypes to production systems.

Human-Robot Interaction

Human-centered analysis of industrial scenes: pose forecasting, safety and cooperation between people and robots on the factory floor.

Computer Vision for Surgery

Applying vision and deep learning to robotic-assisted surgery within the ALTAIR laboratory of the University of Verona.

Efficient Deep Learning

Network interpretability and split computing: understanding where deep networks can be divided to run efficiently on the edge.

EXPERIENCE

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| 2024 — present | Postdoctoral Researcher
<i>University of Verona — ALTAIR Lab & Intelligo Labs</i>
Computer vision and deep learning applied to surgical robotics. |
| Oct 2024 — Dec 2025 | Assistant Researcher
<i>University of Verona</i>
Novel algorithms for robotic-assisted medical surgery, from the computer vision perspective. |
| Feb 2020 — Oct 2021 | AI Software Engineer
<i>Humatics s.r.l., Verona</i>
Computer vision algorithms for industrial applications, in particular surveillance video analysis. |

EDUCATION

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| 2021 — 2024 | PhD in Computer Science
<i>University of Verona</i>
Human-centered scene analysis in industrial environments. Visiting researcher at the University of Glasgow (6 months) and Leibniz University Hannover (3 months). |
| 2017 — 2019 | Master's Degree in Computer Science and Engineering
<i>University of Verona</i>
Curriculum Visual Computing. Thesis on deep learning for 6-DoF object pose estimation in industrial scenarios. |
| 2014 — 2017 | |

Bachelor's Degree in Bioinformatics

University of Verona

Computer science, statistics and biology.

2020

State Exam – Information Engineer (Sez. A)

Italy

Qualification to practice as an Information Engineer, with full marks.

SELECTED PUBLICATIONS

1. 6D Pose Estimation for Industrial Applications

Cunico F., Carletti M., Cristani M., Masci F., Conigliaro D.

International Conference on Image Analysis and Processing (ICIAP) · 2019

2. I-SPLIT: Deep Network Interpretability for Split Computing

Cunico F., Capogrosso L., Setti F., Carra D., Fummi F., Cristani M.

International Conference on Pattern Recognition (ICPR) · 2022

3. Pose Forecasting in Industrial Human-Robot Collaboration

Sampieri A., D'Amely di Melendugno G. M., Avogaro A., Cunico F., Setti F., Skenderi G., Cristani M., Galasso F.

European Conference on Computer Vision (ECCV) · 2022

Full list on Google Scholar: <https://scholar.google.it/citations?user=fvOYgyAAAAAJ>